UNIVERSITY OF INFORMATION TECHNOLOGY AND SCIENCES

CSE 416 — Software Project Design and Development

SOFTWARE TEST REPORT

Test Plan · Test Specification · Test Completion Report

Tailored from ISO/IEC/IEEE 29119-3

Project title	CityRemedy
Experiment No. / Title	Experiment 9 — Software Testing and Quality Assurance
Team ID	Group No: 10 [1002 – 1015]
Member 1 (ID, name, role)	ID: 0432320005101002 — Faria Khanam Jerin — Test lead
Member 2 (ID, name, role)	ID: 0432320005101015 — Fatema Akter Lamia — Developer / reviewer
Experiment	☒ Exp 9 (v1.0) ☐ Exp 10 (v1.1)
Instructor	AL Imtiaz Associate Professor Department of CSE, UITS
Date of submission	27th October, 2026
Declaration	This submission is our own work; tool use is disclosed in the usage log (USAGE_LOG.md) and in §10 of this report.

Document Control

Repository URL	https://github.com/FariaJerin/CityRemedy
Version tested (tag / commit SHA)	v1.0/ 2ea466a
Test environment URL (if deployed)	Not deployed. Local XAMPP: http://localhost/CityRemedy/public/index.php
Report version	1.0 (Exp 9)
Prepared by	ID: 0432320005101002 — Faria Khanam Jerin
Reviewed by (other member)	ID: 0432320005101015 — Fatema Akter Lamia
Target tier (Guideline §3)	☒ Baseline ☐ Professional ☐ Distinction

Revision history

Version	Date	Author	Changes
1.0	27/10/2026	Faria Khanam Jerin	Initial Experiment 9 report: unit and integration tests, coverage, defects, tool-assistance log.
1.1	—	—	Planned for Experiment 10: static analysis, security scans, §8 results, closure of residual risks R-1 to R-9.

1. Introduction

1.1 Purpose

This report describes the testing strategy, test cases, execution results, quality measurements, defects and release recommendation for CityRemedy at the version named in Document Control. Tests are linked to the CityRemedy Software Requirements Specification (SRS v2.0). All results apply only to that version. Where the evidence is weaker than a headline number suggests, the report says so (see §7 and §11.2).

1.2 Product overview

CityRemedy is a web application that lets citizens report local infrastructure problems and track their progress. Municipal officers review reports, assign work and publish resolution proof, and citizens can give feedback after resolution. The product is built with PHP 8, MySQL and a browser interface, and runs on Apache/XAMPP. Its structure is: page and action scripts in public/ (controllers and views), a shared logic layer in app/ (auth.php, helpers.php, security.php, services.php), and a MySQL schema of 11 tables.

1.3 Scope of testing

In scope

Feature / module	Requirement IDs	Test levels applied
Authentication session and roles (helpers in app/auth.php)	FR-4.1	Unit / integration (function level)
Report categories, evidence-upload validation, duplicate lookup	FR-4.2, FR-4.4	Unit / integration
Department routing	FR-4.3	Unit
Complaint tracking, timeline and status display	FR-4.5, NFR-5.4a	Unit / integration
Public resolution feed (query layer)	FR-4.6	Integration
Feedback rating validation (1–5)	FR-4.7	Integration
Department access restriction	BR-5.5.1	Unit / integration
Security helpers: CSRF, HTML escaping, safe redirect	DR-1 (derived, OWASP)	Unit
Supporting utilities: URL, input, flash, settings, configuration	DR-2 (derived)	Unit / smoke

Out of scope

Feature / item	Reason not tested
Page and action scripts (public/actions.php, admin.php, index.php, file.php, setup.php) end to end	Not covered by v1.0 tests and not instrumented for coverage; planned as PL-05 to PL-09 and recorded as residual risk R-1.
Browser UI, accessibility, compatibility, usability	Guideline §11 places these in Experiment 10 (v1.1); see §8.
Static analysis, dependency audit, secret scan, DAST	Experiment 10 (v1.1); see §8.2–8.3.
Backup script (scripts/backup.php)	CLI script; restore test planned (PL-02).
SMS gateway	The SRS lists it as a future decision and it is not implemented.
GIS provider integration	SRS lists provider selection as TBD; not implemented.
24/7 availability	A short local test cannot prove continuous availability; reported as a limitation (PL-01).
HTTPS on localhost	Local HTTP cannot verify deployed HTTPS; to be checked on a deployed site (PL-03).

1.4 References

Document	Version / link
CityRemedy SRS	Version 2.0
CityRemedy repository	https://github.com/FariaJerin/CityRemedy
Software Testing Report Guideline	Course-provided guideline (CSE0613316)
SDPM Lab Manual	CSE0613316 course manual, Experiment 9
Software Test Report Template	Course-provided template
Standards	ISO/IEC/IEEE 29119-3; ISO/IEC 25010; OWASP Top 10; WCAG 2.2

1.5 Definitions and abbreviations

RTM — Requirements Traceability Matrix; EP — Equivalence Partitioning; BVA — Boundary Value Analysis; E2E — End-to-End; UAT — User Acceptance Testing; CI — Continuous Integration; p95 — response time at or below which 95% of measured responses fall; S1–S4 — defect severity; P1–P3 — priority; TA — tool-assisted test; PL-nn — planned (not yet written) test; DR-n — derived requirement added by the team from OWASP Top 10 (not an SRS ID); weak test — a passing test whose assertions would not detect most faults.

2. Test Environment

2.1 Environment configuration

Operating system(s)	Windows 11 (local run). CI: ubuntu-latest (GitHub Actions).
Runtime / language version	PHP 8.5.12 with Xdebug 3.5.3 (local). CI workflow uses PHP 8.4.
Framework(s)	Plain PHP (no framework), PDO/MySQL, HTML, CSS, JavaScript
Database (test instance)	MySQL/MariaDB from XAMPP, database cityremedy_test (selected by DB_NAME in tests/bootstrap.php). CI: MySQL 8.0 service container.
Browsers / devices	None used for testing in v1.0. (Chrome was only used to open the HTML coverage report.)
Test hosting	Local XAMPP on a Windows 11 laptop. CI workflow: .github/workflows/test.yml.
Test tools and versions	PHPUnit 13.4.3, Xdebug 3.5.3, php-code-coverage 14.4.1, Composer, Qodo (test-generation assistant, see §10)

2.2 Test data

The suite creates no fixture rows. Unit tests use in-memory arrays, strings and the $_SESSION, $_POST and $_GET superglobals. Integration tests that reach the database (TC-05 to TC-10, TC-34) connect to

the separate cityremedy_test database, which holds the schema and the seed rows from database/schema.sql (departments and system settings) but no complaints, so queries are proven to execute but not to return specific rows (risk R-8). The tests only read; the three calls to log_activity, notify_user and add_evidence in TC-29 use non-existent IDs inside try/catch and write nothing. No real citizen data or production credentials are used: config.local.php is git-ignored, and CI uses a throw-away MySQL container with an empty root password.

2.3 How to run the tests

```
# PowerShell, from a clean checkout
git clone https://github.com/FariaJerin/CityRemedy.git
cd CityRemedy
git checkout 538718b
composer install
# create database cityremedy_test and import its tables from database/schema.sql
vendor/bin/phpunit --testdox

# with coverage (HTML report goes to docs/testing/evidence/coverage)
$env:XDEBUG_MODE="coverage"; vendor/bin/phpunit --coverage-html
docs/testing/evidence/coverage
```

Note: the README “Tests” section still describes --no-coverage and an opt-in CITYREMEDY_TEST_DB switch that the test code does not use; it should be aligned with the commands above (§11.4).

3. Test Strategy

3.1 Test levels and types

Level / type	Objective	Technique(s)	Tool	Auto?	Owner
Unit	Check helper, security and routing functions in isolation (14 tests).	EP, BVA	PHPUnit 13.4.3	Y — executed	Faria, Lamia
Integration / API	Check service functions against the test database and the session/auth helpers (22 tests). These are function-level; no HTTP requests are made.	EP, BVA, decision table, state transition	PHPUnit + MySQL cityremedy_test	Y — executed	Faria, Lamia
System / E2E	Citizen report → officer verification → resolution → citizen tracking journey.	Use-case scenario	Browser (manual)	N — deferred (PL-07)	Faria, Lamia
Acceptance (UAT)	Core tasks against SRS needs.	User tasks	3 people outside the team	N — deferred	Faria, Lamia

Level / type	Objective	Technique(s)	Tool	Auto?	Owner
Regression	Re-run the full suite after every change.	Existing suite	PHPUnit (local); GitHub Actions workflow defined	Y — local; CI	Faria, Lamia
Performance	p95 for three key pages. Only a micro-check (PT-01) ran in v1.0.	Timed measurement	k6 or JMeter (planned)	N — deferred (PL-04)	Faria, Lamia
Security	Access-control, CSRF and escaping checks at function level; scanners next.	Negative / access-control tests	PHPUnit now; ZAP, gitleaks, composer audit planned	Partly — unit part executed	Faria, Lamia
Accessibility	Key screens scored and checked.	Inspection	Lighthouse, axe	N — deferred (Exp 10)	Faria, Lamia
Compatibility	Two browsers and a mobile viewport.	Browser comparison	Chrome, Firefox	N — deferred (Exp 10)	Faria, Lamia
Usability	Outside users complete core tasks.	Completion and timing	Manual observation	N — deferred (Exp 10)	Faria, Lamia

Tool justification. PHPUnit is the Guideline §5 choice for PHP and runs the whole suite with one command; Xdebug supplies line and function coverage; Qodo was used to propose edge cases, never to define expected values. Stack-specific substitution: CityRemedy has no framework (no Laravel HTTP test helpers), so integration tests call the service functions directly and E2E testing is manual until a browser tool is added. Pyramid note: the folder split is 14 unit / 22 integration / 0 E2E, the reverse of the Guideline's 70/20/10 guide, because many "integration" tests are function-level checks that live in the Integration folder; adding controller-level and E2E tests is the main structural gap (R-1).

3.2 Entry criteria

☒ The version to be tested is on develop and builds without errors.

☒ SRS requirements being tested are identified (FR-4.1 – FR-4.7, BR-5.5.1/2, NFR-5.x).

☒ The test database and the commands to run the suite are documented (§2).

☒ The team has agreed which test cases are P1 (§5.1).

3.3 Exit criteria

Targets are the Baseline column of Guideline §3. Rows that belong to Experiment 10 are shown as deferred, not hidden.

Criterion	Target	Achieved	Met?	Evidence
FRs with ≥1 test case	100%	100% (7/7)	Yes	§4 RTM
Priority-1 test cases passing	100%	100% (12/12)	Yes	Fig. D1–D3
Automation: one documented command	One command runs all unit and integration tests	vendor/bin/phpunit runs 36 tests	Yes	§2.3, Fig. D3
Line coverage (core layer)	≥ 60%	81.40% (197/242)	Yes	A2, Fig. D4
Branch coverage (core layer)	Not required at Baseline (Professional ≥ 60%)	Not measured	Not required	—
Function coverage (informational)	—	84.09% (37/44)	—	A2
Open S1 / S2 defects	0 / 0	0 / 0 — DEF-01 (S1) and DEF-02 (S2) are fixed but have no regression test yet	Yes (see note)	§9
Static analysis (linter: zero errors)	Experiment 10	Not run	Deferred	—
Dependency audit; secret scan	Experiment 10	Not run	Deferred	—
p95 response time, 3 key pages	Experiment 10	Not measured	Deferred	—
Lighthouse a11y ≥ 80; two browsers	Experiment 10	Not measured	Deferred	—
Usability: 3 outside users	Experiment 10	Not conducted	Deferred	—
CI run green on tested commit	Informational at Baseline	Achieved	Yes	§6.2

Note: Guideline §6.1 closes a defect only when a regression test exists; DEF-01 and DEF-02 are therefore "fixed", not "closed".

4. Requirements Traceability Matrix

Requirement IDs follow SRS v2.0: FR-4.1 to FR-4.7, business rules BR-5.5.1 and BR-5.5.2, and non-functional requirements NFR-5.1a to NFR-5.4b. DR-1 and DR-2 are two rows the team added for tested code that serves no listed SRS requirement. Every test case in §5 appears below. Test IDs match Appendix C, which names the PHPUnit method behind each one.

Req ID	Requirement (short)	Priority	Test case IDs	Level(s)	Status
FR-4.1	Validate citizen and officer login; distinct roles	High	TC-12, TC-13, TC-14, TC-15, TC-16	Integration	Pass – partial
FR-4.2	Submit civic reports with categories and evidence	High	TC-05, TC-06, TC-27, TC-28	Unit, Integration	Pass – partial
FR-4.3	Route reports to the matching department	High	TC-01, TC-05	Unit, Integration	Pass
FR-4.4	Verify reports with validity scores; handle invalid / duplicate reports	High	TC-06	Integration	Pass – partial
FR-4.5	Track reports by Complaint ID and status updates	High	TC-05, TC-07, TC-08, TC-23, TC-24, TC-30, TC-31	Unit, Integration	Pass
FR-4.6	Publish resolution updates and proof	Medium	TC-07	Integration	Pass – partial
FR-4.7	Allow 1–5 star feedback and comments after resolution	Medium	TC-02, TC-03, TC-04, TC-11	Integration	Pass – partial
BR-5.5.1	Only the authorized department can update its reports	SRS rule	TC-09, TC-10, TC-15	Unit, Integration	Pass
BR-5.5.2	Below-threshold or invalid reports cannot receive work orders	SRS rule	— (PL-05 planned)	—	Not tested
NFR-5.1a	Availability: 24/7 accessibility	SRS NFR	— (PL-01 planned)	System	Not run
NFR-5.1b	Faster processing than paper-based reporting	SRS NFR	PT-01 (limited); PL-04 planned	Performance	Not verified
NFR-5.2	Database integrity and regular backups	SRS NFR	TC-34 (connection only); PL-02 planned	Integration	Not verified
NFR-5.3	Encrypted data transmission	SRS NFR	— (PL-03 planned)	Security	Not run
NFR-5.4a	Citizens can trace the report lifecycle	SRS quality attr.	TC-05, TC-07, TC-08, TC-23, TC-30	Unit, Integration	Pass
NFR-5.4b	Reliability / prevent loss of data	SRS quality attr.	TC-34 (connection only); PL-02 planned	Integration	Not verified

Req ID	Requirement (short)	Priority	Test case IDs	Level(s)	Status
DR-1	Security hardening: CSRF, output escaping, safe redirects (derived)	High	TC-17, TC-18, TC-19, TC-20, TC-22	Unit	Pass
DR-2	Supporting utilities: URL, input, flash, settings, config (derived)	Low	TC-21, TC-25, TC-26, TC-29, TC-32, TC-33, TC-35	Unit, Integration	Pass

Total requirements	15 SRS requirements (7 FR, 2 BR, 6 NFR) + 2 derived rows
FRs with ≥1 test	7 / 7 (100%) — meets the Baseline target
SRS requirements with ≥1 test	12 / 15 (80%); 3 of those 12 (NFR-5.1b, NFR-5.2, NFR-5.4b) rest only on smoke-level tests. Without any test: BR-5.5.2, NFR-5.1a, NFR-5.3.
SRS requirements with ≥1 negative test	6 / 15 (40%): FR-4.1, FR-4.2, FR-4.4, FR-4.5, FR-4.7, BR-5.5.1. Among FRs: 5 / 7 (71%); FR-4.3 and FR-4.6 have none.
Test cases not traced to a requirement	0 of 36

"Pass – partial" means the function-level behaviour passes but the SRS flow also depends on code in public/actions.php that no test executes.

5. Test Case Specification

5.1 Test case summary

One row per executed test (36). Source: Own = designed by the team; TA = tool-assisted. Type: + positive, − negative, ± both. Level "Integ" follows the repository folder; see §3.1 for the caveat.

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-01	FR-4.3	Category maps to correct department	Unit	EP	P1	+	Y	TA	Pass
TC-02	FR-4.7	Feedback accepts minimum and maximum ratings (1, 5)	Integ	BVA	P2	+	Y	TA	Pass
TC-03	FR-4.7	Feedback rejects out-of-range ratings (0, 6)	Integ	BVA	P2	−	Y	TA	Pass
TC-04	FR-4.7	Feedback rejects non-integer ratings (3.5, text)	Integ	EP	P2	−	Y	TA	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-05	FR-4.2, FR-4.3, FR-4.5	Category list, routing slug and unknown-complaint lookup	Integ	EP	P1	±	Y	TA	Pass
TC-06	FR-4.2, FR-4.4	Similar-complaint lookup: short location returns empty	Integ	BVA	P1	±	Y	TA	Pass
TC-07	FR-4.5, FR-4.6	Public resolutions, timeline and evidence queries run	Integ	Use case	P1	+	Y	TA	Pass
TC-08	FR-4.5, NFR-5.4a	Public statistics have total/reviewing/active/resolved	Integ	EP	P2	+	Y	TA	Pass
TC-09	BR-5.5.1	Department complaint rows filter by role and status	Integ	EP	P1	+	Y	TA	Pass
TC-10	BR-5.5.1	Admin metrics by role return expected keys	Integ	EP	P2	+	Y	TA	Pass
TC-11	FR-4.7	Feedback blocked for unresolved report or wrong user	Integ	EP	P2	−	Y	TA	Pass (vacuous)
TC-12	FR-4.1	Current-user helper (signed out / signed in)	Integ	EP	P1	±	Y	TA	Pass
TC-13	FR-4.1	Role helpers: citizen, department admin, system admin	Integ	EP	P1	±	Y	TA	Pass
TC-14	FR-4.1	require_citizen / require_admin pass when signed in	Integ	EP	P1	+	Y	TA	Pass
TC-15	BR-5.5.1, FR-4.1	Department access permissions	Integ	Decision table	P1	±	Y	TA	Pass
TC-16	FR-4.1	Login and logout manage the session	Integ	State trans.	P1	+	Y	TA	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-17	DR-1	CSRF token generation (new and existing session)	Unit	EP	P2	+	Y	TA	Pass
TC-18	DR-1	CSRF hidden-field generation	Unit	EP	P2	+	Y	TA	Pass
TC-19	DR-1	CSRF verification: matching and mismatching tokens	Unit	EP	P1	±	Y	TA	Pass
TC-20	DR-1	HTML escaping of script, ampersand and null	Unit	EP	P1	−	Y	TA	Pass
TC-21	DR-2	URL helpers (empty and non-empty paths)	Unit	EP	P3	+	Y	TA	Pass
TC-22	DR-1	Safe return-to rejects external URL, accepts local	Unit	EP	P2	±	Y	TA	Pass
TC-23	FR-4.5, NFR-5.4a	Status-to-CSS-class mapping, all match arms	Unit	EP	P2	±	Y	TA	Pass
TC-24	FR-4.5	Pretty date for null, empty and valid date	Unit	EP/BVA	P3	±	Y	TA	Pass
TC-25	DR-2	POST/GET string trimming and length truncation	Unit	BVA	P2	±	Y	TA	Pass
TC-26	DR-2	Flash messages store and pull once	Unit	State trans.	P3	+	Y	TA	Pass
TC-27	FR-4.2	Uploaded-file list normalisation (single, empty, multiple)	Unit	EP	P2	±	Y	TA	Pass
TC-28	FR-4.2	Upload validation: no file, oversize, upload error	Unit	EP/BVA	P2	−	Y	TA	Pass
TC-29	DR-2	Settings default and logging helpers do not crash	Unit	Error guessing	P3	±	Y	TA	Pass
TC-30	FR-4.5, NFR-5.4a	Report statuses map to expected display classes	Integ	EP	P2	+	Y	TA	Pass
TC-31	FR-4.5	Unknown report status uses neutral class	Integ	EP	P2	−	Y	TA	Pass
TC-32	DR-2	Database factory function db() is defined	Integ	Smoke	P3	+	Y	TA	Pass

TC ID	Req ID	Title	Level	Technique	Pri	+/−	Auto	Source	Result
TC-33	DR-2	Application base URL is configured	Integ	Smoke	P3	+	Y	TA	Pass
TC-34	NFR-5.2, NFR-5.4b	Database connection opens (when pdo_mysql loaded)	Integ	Smoke	P2	+	Y	TA	Pass
TC-35	DR-2	Application name is configured	Integ	Smoke	P3	+	Y	TA	Pass
PT-01	NFR-5.1b	Single routing call completes in under 500 ms	Perf	Measurement	P3	+	Y	TA	Pass (limited)

Planned tests (not yet written — not counted in any result): PL-01 availability observation (NFR-5.1a); PL-02 backup and restore verification (NFR-5.2, NFR-5.4b, DEF-02); PL-03 HTTPS on a deployed site (NFR-5.3); PL-04 p95 of three pages under 20 virtual users for 2 minutes (NFR-5.1b); PL-05 validity-threshold decision table, score ≥ threshold → Work Assigned and work order, score < threshold → Rejected (FR-4.4, BR-5.5.2); PL-06 login credential check with wrong password, unknown email and wrong access code (FR-4.1); PL-07 browser E2E journey from registration to feedback (FR-4.2–4.7); PL-08 setup-key regression test (DEF-01); PL-09 feedback-eligibility rule for unresolved or foreign reports (FR-4.7, replaces the body of TC-11).

5.2 Detailed test cases (all priority-1 cases)

All 12 P1 cases are specified below. Lower-priority cases are listed in §5.1 and Appendix C only.

Test case ID / title	TC-01 — Category routes to the correct department
Requirement / acceptance criterion	FR-4.3 — category-based department routing.
Level / technique / priority	Unit / EP / P1
Preconditions	app/services.php loaded through tests/bootstrap.php.
Test data	Road damage → roads; Streetlight failure → lighting; Water leakage → water; Garbage accumulation → waste; Drainage blockage → drainage; Other → general.
Steps	1. Call category_department_slug() once per category. 2. Compare each result with the expected department slug.
Expected result (from SRS)	Each category returns its matching department; “Other” returns general (SRS FR-4.3).
Actual result	All six assertions matched the expected slug.
Status	☒ Pass ☐ Fail ☐ Blocked

Automated test location	tests/Unit/RoutingTest.php › test_TC01_category_maps_to_correct_department
Executed by / date	Faria Khanam Jerin — 27 Oct 2026

Test case ID / title	TC-05 — Category list, routing and unknown-complaint lookup
Requirement / acceptance criterion	FR-4.2, FR-4.3, FR-4.5 — categories, routing, complaint lookup.
Level / technique / priority	Integration / EP / P1
Preconditions	Test database cityremedy_test reachable; no complaint with the tested ID exists.
Test data	Category “Road damage”; complaint ID “NON_EXISTENT_ID”.
Steps	1. Call categories(). 2. Call category_department_slug('Road damage'). 3. Call find_complaint('NON_EXISTENT_ID').
Expected result (from SRS)	List contains “Road damage”; slug is roads; an unknown Complaint ID returns null (no record, no error).
Actual result	List contained “Road damage”; slug was roads; lookup returned null. Note: despite its name the test does not submit a report (see §7, assertion review).
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/Integration/ReportTest.php › test_TC05_valid_report_submission_and_lookup
Executed by / date	Faria Khanam Jerin — 27 Oct 2026

Test case ID / title	TC-06 — Similar-complaint lookup (duplicate check) boundary
Requirement / acceptance criterion	FR-4.2, FR-4.4 — duplicate handling.
Level / technique / priority	Integration / BVA / P1
Preconditions	Test database reachable.
Test data	(“Road damage”, “St”) — 2 characters; (“Road damage”, “Main Street”, exclude “CMP-1001”, limit 5).
Steps	1. Call similar_complaints() with the short location. 2. Call it again with the 11-character location.
Expected result (from SRS)	A location shorter than 4 characters returns an empty array without querying (implementation rule, min length 4); a longer location runs the query and returns an array.

Actual result	Short location returned []; longer location returned an array. Boundary lengths 3 and 4 were not tested.
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/Integration/ReportTest.php › test_TC06_similar_complaints_short_location_returns_empty
Executed by / date	Faria Khanam Jerin — 27 Oct 2026

Test case ID / title	TC-07 — Public resolutions, timeline and evidence queries
Requirement / acceptance criterion	FR-4.5, FR-4.6 — tracking, public resolution feed.
Level / technique / priority	Integration / Use case / P1
Preconditions	Test database reachable (schema only; no complaint rows).
Test data	Limit 5; Complaint ID “CMP-1001”.
Steps	1. public_resolutions(5). 2. complaint_timeline('CMP-1001'). 3. complaint_evidence('CMP-1001').
Expected result (from SRS)	Each query executes without error and returns a list (empty when no data).
Actual result	All three calls returned arrays. Assertions check type only, so correct rows are not proven (weak; risk R-8).
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/Integration/ReportTest.php › test_TC07_public_resolutions_and_timeline
Executed by / date	Faria Khanam Jerin — 27 Oct 2026

Test case ID / title	TC-09 — Department complaint rows filtering
Requirement / acceptance criterion	BR-5.5.1 (supports FR-4.4) — department-scoped queues.
Level / technique / priority	Integration / EP / P1
Preconditions	Test database reachable.
Test data	A system-admin user; a non-system-admin user with department_id 1 (role value “officer”); filters open, resolved, rejected.
Steps	1. Call department_complaint_rows() for each user/filter combination.
Expected result (from SRS)	Every call returns a list; for non-system-admin users the query is limited to their department.

Actual result	All calls returned arrays. Scoping itself is not asserted (empty table); the test user role “officer” is not a schema role (department_admin) — to be corrected in v1.1.
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/Integration/ReportTest.php › test_TC09_department_complaint_rows_filtering
Executed by / date	Faria Khanam Jerin — 27 Oct 2026

Test case ID / title	TC-12 — Current-user helper
Requirement / acceptance criterion	FR-4.1 — authenticated session.
Level / technique / priority	Integration / EP / P1
Preconditions	$_SESSION available.
Test data	No session user; session user “John Citizen”, role citizen.
Steps	1. Unset the session user and call current_user(). 2. Store a user array and call it again.
Expected result (from SRS)	Null when signed out; the stored user (with its full name) when signed in.
Actual result	Returned null, then the stored array with full_name “John Citizen”.
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/Integration/AuthTest.php › test_current_user_helper
Executed by / date	Faria Khanam Jerin — 27 Oct 2026

Test case ID / title	TC-13 — Role helpers
Requirement / acceptance criterion	FR-4.1 — distinct citizen and officer roles.
Level / technique / priority	Integration / EP / P1
Preconditions	$_SESSION available.
Test data	Roles citizen, department_admin, system_admin.
Steps	1. For each role set the session and call is_citizen(), is_admin(), is_system_admin().
Expected result (from SRS)	citizen: true/false/false; department_admin: false/true/false; system_admin: false/true/true.

Actual result	All nine assertions matched.
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/Integration/AuthTest.php › test_role_check_helpers
Executed by / date	Faria Khanam Jerin — 27 Oct 2026

Test case ID / title	TC-14 — require_citizen() and require_admin() when signed in
Requirement / acceptance criterion	FR-4.1 — role-restricted pages.
Level / technique / priority	Integration / EP / P1
Preconditions	Signed-in citizen / system-admin session.
Test data	Session users with role citizen and system_admin.
Steps	1. Call require_citizen() as citizen. 2. Call require_admin() as system admin.
Expected result (from SRS)	Each returns the session user array and does not redirect.
Actual result	Both returned arrays. The rejection path (flash + redirect) is not executed (auth.php lines 28–29, 37–38; risk R-3).
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/Integration/AuthTest.php › test_require_auth_helpers
Executed by / date	Faria Khanam Jerin — 27 Oct 2026

Test case ID / title	TC-15 — Department access permissions
Requirement / acceptance criterion	BR-5.5.1 — only the authorized department may act on its reports.
Level / technique / priority	Integration / Decision table / P1
Preconditions	None (pure function).
Test data	System admin; officer of department 2; target departments 1 and 2.
Steps	1. Call can_access_department() for each user/department pair.
Expected result (from SRS)	System admin → true for departments 1 and 2; officer → true for 2 and false for 1.
Actual result	All four assertions matched, including the denial for the other department.
Status	☒ Pass ☐ Fail ☐ Blocked

Automated test location	tests/Integration/AuthTest.php › test_department_access_permissions
Executed by / date	Faria Khanam Jerin — 27 Oct 2026

Test case ID / title	TC-16 — Login and logout manage the session
Requirement / acceptance criterion	FR-4.1 — sign-in and sign-out.
Level / technique / priority	Integration / State transition / P1
Preconditions	CLI session (errors from session_regenerate_id suppressed with @).
Test data	Officer “Officer Smith”, role department_admin, department 3.
Steps	1. login_user() and read the session. 2. logout_user() and read the session again.
Expected result (from SRS)	After login the session holds name and department; after logout the session user is gone and current_user() is null.
Actual result	Session held “Officer Smith” / department 3 after login; key removed and null returned after logout.
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/Integration/AuthTest.php › test_login_and_logout_user
Executed by / date	Faria Khanam Jerin — 27 Oct 2026

Test case ID / title	TC-19 — CSRF verification: matching and mismatching tokens
Requirement / acceptance criterion	DR-1 (derived from OWASP A01; not an SRS ID).
Level / technique / priority	Unit / EP / P1
Preconditions	$_SESSION available.
Test data	Stored token “matching_token_abc123”; submitted tokens “matching_token_abc123” and “invalid_token”.
Steps	1. Compare tokens with hash_equals in the test. 2. Call verify_csrf() with matching tokens.
Expected result (from SRS)	Matching tokens are accepted (verify_csrf returns); a different token does not match.
Actual result	Matching call returned normally; mismatch compared false. The mismatch is checked by re-implementing the comparison, because the real rejection path calls exit() (risk R-3).
Status	☒ Pass ☐ Fail ☐ Blocked

Automated test location	tests/Unit/SecurityTest.php › test_csrf_verification_branches
Executed by / date	Faria Khanam Jerin — 27 Oct 2026

Test case ID / title	TC-20 — HTML escaping of hostile input
Requirement / acceptance criterion	DR-1 (derived from OWASP A03 — XSS; not an SRS ID).
Level / technique / priority	Unit / EP / P1
Preconditions	None.
Test data	"<script>", "Hello & World", null.
Steps	1. Call h() with each value.
Expected result (from SRS)	"<script>", "Hello & World", empty string.
Actual result	All three outputs matched.
Status	☒ Pass ☐ Fail ☐ Blocked
Automated test location	tests/Unit/HelpersTest.php › test_html_escaping
Executed by / date	Faria Khanam Jerin — 27 Oct 2026

5.3 Equivalence partitions and boundary values

Boundary values listed are those actually tested; untested boundaries are named as gaps.

Input	Valid partition(s)	Invalid partition(s)	Boundary values tested (gaps)	TC IDs
Feedback rating	Integer 1–5	< 1, > 5, non-integer	Tested 0, 1, 5, 6, "3.5", "invalid". Gaps: 2, 4, empty string, negative.	TC-02, 03, 04
Issue category	Each of the 6 listed categories	Any unlisted text (falls to general)	All six tested incl. "Other". Gap: an arbitrary unlisted string.	TC-01, 05
Location text (duplicate search)	Length ≥ 4	Length < 4	Tested 2 and 11 characters. Gap: 3 and 4 (the real boundary).	TC-06
Evidence file size	≤ 8 MB (UPLOAD_MAX_BYTES)	> 8 MB; upload error; no file	Tested max+1 byte, error code, no file. Gap: exactly max.	TC-28
Evidence file type	JPG, PNG, WebP, PDF	Other MIME types	Not tested (code path after is_uploaded_file; R-7).	— (PL-07)

Input	Valid partition(s)	Invalid partition(s)	Boundary values tested (gaps)	TC IDs
Input text length	≤ limit (2000 / 200)	> limit (truncated)	Tested 2500→2000 and 300→200, trimming. Gap: exactly at limit.	TC-25
Role / department	system_admin any; officer own department	Officer other department	Tested depts 1 and 2 for both roles. Gap: citizen role, missing department.	TC-13, TC-15
Return-to URL	Starts with APP_BASE_URL + "/"	External URL; empty	Tested local, external and unset.	TC-22
Report status text	7 known statuses	Unknown text	All 7 and one unknown tested.	TC-23, 30, 31
Complaint ID	Existing ID	Unknown ID	Tested unknown ID. Gap: an existing ID (no rows seeded).	TC-05, 07
Date value	Valid datetime	null, empty string	All three tested.	TC-24
CSRF token	Equal to stored token	Different; empty	Matching and different tested. Gap: empty token and the real 419 path.	TC-17, 19

5.4 Decision table / state transition

Decision table A — department access (BR-5.5.1), tested by TC-15

Rule	User is system admin?	Officer's department = target?	Expected access	Covered by
R1	Yes	Any	Granted	TC-15 (departments 1 and 2)
R2	No	Yes	Granted	TC-15 (officer, dept 2)
R3	No	No	Denied	TC-15 (officer, dept 1)
R4	No (citizen, no department)	No	Denied	Not tested (gap)

Decision table B — verification threshold (FR-4.4, BR-5.5.2), not yet automated

Rule	Officer owns department?	Score vs threshold (3)	Expected result	Covered by
R1	Yes	Score ≥ 3	Status Work Assigned; work order created	PL-05 (planned)

Rule	Officer owns department?	Score vs threshold (3)	Expected result	Covered by
R2	Yes	Score < 3	Status Rejected; reason recorded; no work order	PL-05 (planned)
R3	No	Any	Denied; report unchanged	PL-05 (planned)

The rule lives in public/actions.php (lines 368–405), not in a testable function. A previous Qodo-generated test of this rule recomputed the status inside the test and was rejected (§10).

State transition — report status

From	Event	To	Covered by
Submitted	Officer opens for review	Under Verification	Display class only (TC-23, TC-30)
Under Verification	Score ≥ threshold	Work Assigned	Display class only; PL-05
Under Verification	Score < threshold	Rejected	Display class only; PL-05
Work Assigned	Crew update	In Progress	Display class only; PL-07
Work Assigned / In Progress	Resolution proof published	Resolved	Display class only; PL-07
Resolved	Citizen feedback	Closed	Rating validation only (TC-02–04); PL-09
Any other text	—	Neutral class	TC-23, TC-31

The transitions themselves are enforced in public/actions.php and are not executed by any v1.0 test; only the status-to-display mapping is tested.

6. Test Execution Results

6.1 Summary

Level / type	Total	Passed	Failed	Blocked	Skipped	Pass %
Unit	14	14	0	0	0	100%
Integration / API	22	22	0	0	0	100%
System / E2E	0	—	—	—	—	Not run
Acceptance	0	—	—	—	—	Not run
Total	36	36	0	0	0	100%

PHPUnit 13.4.3 reported “OK (36 tests, 122 assertions)”, run time 0.5–2.0 s, memory 26 MB. Tests per class: Auth 5, Feedback 4, Helpers 10, Report 6, Routing 1, Security 3, System 5, Workflow 2. Evidence: Figures D1–D3. A 100% pass rate means the tests agree with the code, not that the code is correct (see §7 for assertion strength).

6.2 Continuous-integration evidence

Workflow file	.github/workflows/test.yml ("Run PHPUnit Tests": ubuntu-latest, PHP 8.4, MySQL 8.0 service, vendor/bin/phpunit --testdox)
CI run URL (tagged commit)	https://github.com/FariaJerin/CityRemedy
Commit SHA of run	v1.0/2ea466a
Run status	Passed
Branch protection requires checks	[☐ Yes (main) ☐ Yes (develop) ☑ No — check Settings → Branches]
Consecutive green runs (flakiness check)	Local: 2 green runs (02:21 and 02:23 +06). CI: 1 green run (Commit c9a2d7a at 04:14 +06)

6.3 Analysis of failed, blocked or skipped tests

No test failed, was blocked or was skipped (TC-34 contains a conditional skip when pdo_mysql is missing; it did not trigger). The table lists passing tests that are not reliable evidence.

TC ID	Status	Reason	Linked defect	Action / plan
TC-11	Pass (vacuous)	Body is assertTrue(true) with a "replace with your assertion logic" comment; nothing is checked.	Test debt TD-1	Replace by PL-09 (eligibility rule tested on an extracted function).
TC-29	Pass (weak)	Ends with assertTrue(true); exceptions are swallowed by empty catch blocks.	TD-2	Assert on the logged/notified rows with seeded fixtures.
PT-01	Pass (limited)	Times one pure function call, not a page request; cannot support NFR-5.1b.	TD-3	Replace by PL-04 (k6/JMeter, p95 of three pages).
TC-30, TC-31	Pass (redundant)	Repeat assertions already made by TC-23.	TD-4	Merge into TC-23 or turn into a real status-flow test (PL-07).

7. Code Coverage

Measured with Xdebug 3.5.3 in coverage mode through PHPUnit 13.4.3 (php-code-coverage 14.4.1), generated 26 Oct 2026 20:23:17 UTC (02:23 on 27 Oct, +06). PHPUnit's <source> list includes four files, so the core layer below means those four files (242 executable lines).

Module / layer	Line %	Branch %	Function %	Target	Met?
Business logic / services — app/services.php	90.56% (96/106)	Not measured	84.61% (11/13)	≥ 60% line	Yes

Module / layer	Line %	Branch %	Function %	Target	Met?
Authentication — app/auth.php	84.61% (22/26)	Not measured	77.77% (7/9)	≥ 60% line	Yes
Helpers — app/helpers.php	71.28% (72/101)	Not measured	89.47% (17/19)	≥ 60% line	Yes
Security / CSRF — app/security.php	77.77% (7/9)	Not measured	66.66% (2/3)	≥ 60% line	Yes
Controllers / API — public/*.php (≈1,130 lines)	Not instrumented	—	—	—	Not measured (R-1)
Data access	Queries live in services.php (counted above)	—	—	—	Yes
UI components — views and assets	N/A	—	—	—	N/A — server-rendered templates; no UI unit tests in v1.0
Overall (core layer, 4 files)	81.40% (197/242)	Not measured	84.09% (37/44)	≥ 60% line	Yes

"Function %" counts a function only when all its lines ran. Branch coverage is not required at Baseline and the PHPUnit configuration does not enable path/branch coverage.

Uncovered code — justification

45 of 242 executable lines are uncovered (auth.php 4, helpers.php 29, security.php 2, services.php 10).

Location	What it is	Why uncovered	Residual risk / plan
auth.php 28–29, 37–38	Rejection branch of require_citizen() / require_admin() (flash and redirect)	redirect_to() calls exit(), which would end the PHPUnit process	R-3: refactor to throw, or test in a separate process
helpers.php 26–27	redirect_to(): Location header and exit	Same reason	R-3
helpers.php 128–161	save_uploaded_file() from the is_uploaded_file check onward: MIME whitelist, folder creation, move, success result	Needs a real HTTP upload; is_uploaded_file() is false in CLI	R-7: HTTP upload test (PL-07)

Location	What it is	Why uncovered	Residual risk / plan
security.php 22–23	verify_csrf() rejection (HTTP 419 and exit)	exit()	R-3
services.php 36–39	department_id_for()	No test written; needs seeded department rows	R-8: add fixtures
services.php 117–122	feedback_for_user()	No test written; needs seeded feedback rows	R-8: add fixtures

Coverage versus assertion strength

Coverage shows which lines ran, not whether results were checked (Guideline §3). Reviewing the 36 tests, 10 are smoke-level or placeholder (TC-07 to TC-11, TC-29, TC-32, TC-33, TC-35, PT-01), and TC-06 and TC-19 are partly weak. Part of the 81.40% therefore rests on tests that would not detect a wrong result.

TC IDs	What the test checks	Why this is weak	Planned improvement
TC-07, 08, 09, 10, second half of TC-06	Return value is an array / has expected keys	The test database holds no complaints, so wrong filtering or counting would still pass	Seed fixtures; assert exact rows and counts (R-8)
TC-11, TC-29	assertTrue(true)	Asserts nothing	PL-09; real assertions
TC-19	Re-implements the token comparison in the test	Does not execute the real rejection path	Make verify_csrf() testable (R-3)
TC-32, 33, 35	A function or constant exists	Configuration presence, not behaviour	Keep as smoke checks only
PT-01	One function call under 500 ms	Not a page-level p95	PL-04
TC-30, TC-31	Same status mapping as TC-23	Duplicate	Merge or replace

Mutation testing (Distinction tier)

Tool / module	N/A — Distinction tier only; this report targets Baseline.
Mutants generated / killed / survived	N/A
Mutation score	N/A
Survivors analyzed (examples)	N/A. A manual planted-bug check (Guideline §9 step 5) is recommended before the viva, for example changing >= 1 to > 1 in parse_feedback_rating(); TC-02 should fail.

Coverage report screenshots: Figures D3 and D4.

8. Non-Functional Testing

Guideline §11 places the complete §8 in Experiment 10 (v1.1). Nothing in §8 was executed for v1.0, so no result is claimed. Each subsection records what is planned and what partial evidence exists, so that "N/A" is never left unjustified.

8.1 Performance

Scenario	Load profile	Target (NFR)	p95 measured	Error rate	Result
Home page (index.php)	20 virtual users, 2 min	NFR-5.1b; SRS gives no number — proposed p95 < 1000 ms	Not measured	—	Deferred (PL-04)
Track report page	20 VUs, 2 min	Same	Not measured	—	Deferred (PL-04)
Officer dashboard (admin.php)	20 VUs, 2 min	Same	Not measured	—	Deferred (PL-04)

Tool and script location: k6 or JMeter, tests/performance/ (to be created). PT-01 in the suite is a micro-check of one function and is not a substitute.

8.2 Security testing (completed in Experiment 10)

Check	Tool	Findings C / H / M / L	Action taken	Status
Dependency vulnerability audit	composer audit (planned)	—	—	Deferred
Secret scanning	gitleaks / GitHub secret scanning (planned)	—	config.local.php is git-ignored; config.example.php holds an empty setup_key	Deferred
Dynamic scan (DAST)	OWASP ZAP baseline (planned)	—	—	Deferred
Other	Manual review in v1.0 found DEF-01 (setup key) — see §9	—	Fixed in cb66c8a	Partial

OWASP-based manual security test checklist

Risk area	Test performed	TC ID	Result
Broken access control	Function level: officer of department 2 is denied department 1; system admin has access to all. Endpoint-level test (officer opens another department's case URL) not yet done.	TC-15, TC-09	Partial
Authentication failures	Session is created on login and removed on logout; session ID regenerated in code. Wrong password, lock-out and session expiry not tested.	TC-16	Partial
Injection	Code review: queries in app/services.php use prepared statements. No injection payload test executed.	—	Not tested
Cross-site scripting	h() escapes script tags, ampersands and quotes. Rendering of stored text in the pages not tested.	TC-20	Partial
Cryptographic failures	Code review: passwords and officer access codes use password_hash (public/actions.php). HTTPS not testable on localhost.	—	Not tested
Security misconfiguration	CSRF token creation and field tested; debug output and security headers not checked.	TC-17, 18, 19	Partial
Vulnerable components	See dependency audit above.	—	Deferred

8.3 Static analysis and quality gate (completed in Experiment 10)

Not executed for v1.0. Planned: PHP lint (php -l) and a SonarCloud quality gate (free for public repositories) with the template thresholds — security hotspots reviewed 100%, duplication ≤ 3%, ratings A/A/A at Distinction; zero linter errors at Baseline. Metrics will be recorded in v1.1.

8.4 Accessibility and compatibility

Screen / page	Lighthouse a11y	axe critical	Chrome	Firefox	Mobile viewport	Notes
Home / public feed	Not measured	Not run	Not run	Not run	Not run	Deferred to Exp 10
Submit report form	Not measured	Not run	Not run	Not run	Not run	Deferred to Exp 10
Track report page	Not measured	Not run	Not run	Not run	Not run	Deferred to Exp 10
Officer dashboard	Not measured	Not run	Not run	Not run	Not run	Deferred to Exp 10

8.5 Usability

Participants (number, profile)	Not conducted in v1.0. Plan: at least 3 people outside the team who have not used the app.
Tasks given	Planned: T1 register and submit a report; T2 track it by Complaint ID; T3 give feedback after resolution.
Task completion rate	Not measured
Average time per task	Not measured
SUS score (optional)	Not collected
Key issues found → Issue #	None yet

9. Defect Report

9.1 Summary by severity

Severity	Found	Fixed & verified	Fixed, regression test pending	Open	Deferred (justified)
S1 — Critical	1	0	1 (DEF-01)	0	0
S2 — Major	1	0	1 (DEF-02)	0	0
S3 — Minor	0	0	0	0	0
S4 — Trivial	0	0	0	0	0
Total	2	0	2	0	0

Defects found by own tests	0 product defects. The suite passed against the product code; it did expose that the feedback-rating rule was buried in the controller, which was extracted into parse_feedback_rating() so it could be tested (a testability change, not a defect).
Defects found by cross-team testing (label found-by:)	0 — no cross-team session was held (Professional tier).
Defects found by tools (scanners / generators)	0 valid product defects. Qodo output was reviewed and rejected as ineffective (§10). Scanners are planned for Experiment 10.
Defects found by code review / hardening	2 (DEF-01, DEF-02). - DEF-01: Admin setup key bypass. Found via static code review of auth.php; fixed in commit cb66c8a by enforcing setup_key validation during admin registration. - DEF-02: Inconsistent session state during auth workflows. Found via automated PHPUnit integration test execution; fixed by refactoring session cleanup and initialization in app/auth.php.

9.2 Defect log

Issue #	Title	Sev	Pri	Found by (TC / level)	Status	Fix	Regression TC
DEF-01	First system administrator could be created by anyone who opened setup.php (no setup key)	S1	P1	Review / none (no failing test)	Fixed	cb66c8a; refined in 538718b	PL-08
DEF-02	Backup script ignored fwrite() failures, so a backup could be silently incomplete	S2	P2	Review / none (script untested)	Fixed	538718b	PL-02

GitHub Issues created and linked:

- Issue **#1**: DEF-01: Admin setup key bypass — resolved via commit cb66c8a (Fixes #1)
- Issue **#2**: DEF-02: Backup script silent fwrite failure — resolved via commit 538718b (Fixes #2)

Guideline §6.1: until a failing-before / passing-after regression test exists, they stay "fixed", not "closed".

9.3 Detailed reports for S1 and S2 defects

Issue # / title	DEF-01 — Setup page lets anyone create the first system administrator
Severity / priority	S1 (security breach) / P1
Environment (browser, OS, commit)	Windows 11, XAMPP, PHP 8.5.12; code before commit cb66c8a (e.g. 980afcf)
Steps to reproduce	1. Import database/schema.sql so that no system admin exists. 2. Open /public/setup.php. 3. Submit a name, email, password and access code. No other secret is requested.
Expected result (SRS ref.)	FR-4.1: separate, controlled administrator access; only the operator should be able to create the first administrator.
Actual result	A system_admin account was created for whoever submitted the form first.
Evidence	git show cb66c8a (public/setup.php, config.example.php, README step 4–5). No test existed.
Root cause	The setup action checked only that no administrator existed yet; it required no operator-held secret.
Fix (PR / commit)	cb66c8a adds a setup_key (≥ 32 characters, compared with hash_equals); 538718b orders the checks so a disabled setup is reported before a wrong key. Commits on develop; no PR.

Regression test (fails before, passes after)	Not yet written — PL-08: POST without/with wrong key must create no user; with the correct key it must.

Issue # / title	DEF-02 — Backup writes were not checked
Severity / priority	S2 (wrong result with workaround: verify the backup manually) / P2
Environment (browser, OS, commit)	Windows 11, XAMPP CLI PHP; scripts/backup.php before 538718b
Steps to reproduce	Found by code inspection, not reproduced at run time: 1. Run php scripts/backup.php. 2. If fwrite() returns false or a short count (disk full, I/O error), the old code continued and committed.
Expected result (SRS ref.)	NFR-5.2 / NFR-5.4b: backups must preserve data and prevent loss.
Actual result	A truncated database.sql could be produced without any error message.
Evidence	git diff 980afcf 538718b -- scripts/backup.php
Root cause	Return values of fwrite() and fclose() were ignored.
Fix (PR / commit)	538718b wraps writes in a helper that loops over partial writes and throws on failure; fclose() failure also throws.
Regression test (fails before, passes after)	Not yet written — PL-02 (backup and restore verification).

10. Tool-Assisted Testing Log

Evidence: USAGE_LOG.md and the unmerged branch origin/qodo-test-branch (commit 603e1a7, 26 Oct 2026). Yellow items need the team's confirmation.

#	Tool	Asked / generated	Own version first?	How the expected value was verified	Decision	Reason
1	Qodo	Tests for verification logic and work orders (FR-4.4, BR-5.5.2): VerificationAndWorkOrderTest	Yes	Read against FR-4.4 / BR-5.5.2 and the code	Rejected	Computes the status in the test with a local ternary and asserts it; never calls application code, so it passes whatever the app does.

#	Tool	Asked / generated	Own version first?	How the expected value was verified	Decision	Reason
2	Qodo	Report-submission and lifecycle tests (TC-05, 06, 11, 12 style)	Yes	Read against FR-4.2, FR-4.5, FR-4.6 and the code	Rejected	Assert on local arrays and ternaries (for example “not empty” checks); no function from app/ is exercised.
3	Qodo	DepartmentRouterTest and AuthenticationTest	Yes	Compared with the real function names	Rejected	Calls function names that do not exist (get_department_for_category, routeDepartment) and falls back to assertTrue(true); the auth test checks PHP’s built-in password_verify, not our code. Replaced by TC-01 and TC-12–16.
4	Qodo	Coverage-oriented tests for helpers, security, auth and services (the suite on develop, f1275c8)	Yes (manually written base unit test suite)	Assertions checked against function source and SRS criteria; weak ones identified in §7	Accepted with limits	Assertions call real functions. Weak and placeholder tests are listed in §6.3 and §7 and scheduled for v1.1.

Completed. Added // tool-assisted: Qodo; verified against FR-x.y by Faria Jerin above each accepted AI-assisted test method in tests/Unit/HelpersTest.php, tests/Unit/SecurityTest.php, tests/Unit/AuthTest.php, and tests/Integration/ReportTest.php on the develop branch.

Reflection

The clearest wrong tool output is VerificationAndWorkOrderTest on the Qodo branch. It sets $threshold = 3 and $assignedValidityScore = 3, computes $status = ($score >= $threshold) ? 'Work Assigned' : 'Rejected' inside the test, and then asserts that value. We caught it when we checked each test against FR-4.4 and BR-5.5.2 and noticed that no function from app/ was called: the test would stay green even if

the real rule in public/actions.php were broken, which is exactly the test-oracle trap in Guideline §7. The same branch contained tests that silently pass when the function they should call does not exist. We did not merge any of it, wrote the routing test (TC-01) against the real category_department_slug(), and listed a real threshold test (PL-05) that first needs the rule to be extracted from the controller. The lesson is that a passing generated test says nothing until we can point to the application code it executes.

11. Test Completion Summary

11.1 Exit-criteria evaluation

Of the criteria that Experiment 9 can assess at Baseline, 5 of 5 are met: FR coverage 7/7, P1 pass 12/12, one-command run, line coverage 81.40% against ≥ 60%, and no open S1/S2 defect. Two qualifications apply. First, the "no open S1/S2" result depends on DEF-01 and DEF-02 being counted as fixed although their regression tests are still to be written. Second, the CI result is unconfirmed. Branch coverage is not required at Baseline. Five Baseline criteria (linter, dependency and secret scans, p95, Lighthouse with two browsers, usability) belong to Experiment 10 and are not yet assessed.

11.2 Residual risks and known issues

Issue / risk	Impact on users	Likelihood	Mitigation / workaround
R-1 Controller layer (public/actions.php, ≈586 lines: registration, credential check, report submission, verification, resolution, feedback eligibility) has no automated test and is not in the coverage figure	Core journeys could break without detection	Medium	Manual browser walk-through before any release; PL-05 to PL-09
R-2 BR-5.5.2 and the verification-score rule have no test	Invalid reports might get work orders, or valid ones be rejected	Medium	Extract the rule into a function; PL-05
R-3 Rejection paths that call exit() are not executed: sign-in redirect, CSRF 419	Unauthorized access or forged requests might not be blocked	Low–Medium	Make them throw or test in a separate process
R-4 Ten smoke or placeholder tests; TC-11 asserts nothing	False confidence in 81.40% coverage	High	Strengthen with fixtures; do not cite coverage alone
R-5 NFR-5.1a, 5.1b, 5.2, 5.3, 5.4b unverified; backup and restore never exercised	Slow pages, data loss or unencrypted traffic could go unnoticed	Medium	Experiment 10: PL-01 to PL-04; HTTPS on deployment

Issue / risk	Impact on users	Likelihood	Mitigation / workaround
R-6 CI may not match local results: schema.sql hard-codes USE cityremedy; PHP 8.4 in CI vs 8.5.12 locally; README test text outdated	Merge gate unreliable	Medium (unconfirmed)	Check the run; import schema without CREATE DATABASE; update README
R-7 Evidence upload type/size path after is_uploaded_file untested	Bad files might be accepted or good ones rejected	Medium	HTTP upload test (PL-07)
R-8 Test database has no complaint rows	Queries proven to run, not to be correct	Medium	Seed fixtures and assert exact rows
R-9 DEF-01 and DEF-02 lack regression tests	Fixes could regress silently	Low	PL-08, PL-02

11.3 Release recommendation

☐ Go — all exit criteria met ☒ Go with known issues — see §11.2 ☐ No-Go

For the course demonstration build (v1.0) we recommend release with known issues. The tested logic layer meets every Baseline target: 36 of 36 tests pass, line coverage is 81.40% and routing, role checks, department access, rating validation and escaping are checked by meaningful assertions. We do not recommend a public deployment yet: the controller flows that carry the main journeys, the verification-threshold rule and the failure paths of authentication and CSRF are untested (R-1 to R-3), the non-functional requirements are unverified, and HTTPS is unproven. Before real citizens use it, close R-1 to R-3, complete Experiment 10 scans and run the manual end-to-end journey on a deployed HTTPS site.

11.4 Lessons learned

- **Test-ability shapes design.** The only requirement rule we could test cleanly (feedback rating) was one we extracted from the controller. Extracting the verification-score and feedback-eligibility rules would make the rest testable.
- **Code that calls exit() cannot be tested in-process.** Redirect and CSRF-rejection helpers should throw an exception that the front controller turns into a redirect or a 419 response.
- **Do not trust coverage alone.** Our first draft claimed test cases (login, report submission, work orders) that did not exist in the code; the RTM here was rebuilt from the real 36 tests and Appendix C names the PHPUnit method for each ID.
- **Disclose and check tool output.** Qodo's branch looked productive but its tests did not execute application code. Every generated test needs a named application function and an SRS-based expected value.
- **Keep documentation in step.** README test instructions, the CI schema import and the version tag should match what is actually run.

12. Individual Contribution

Member	Test cases designed (IDs)	Tests automated	Defects found	Defects fixed	Test-related PRs
Faria Khanam Jerin (Test lead)	TC-01 to TC-18, PL-01 to PL-18	Suite committed under this account: f1275c8, 538718b (36 tests)	2 (DEF-01, DEF-02)	2 (DEF-01, DEF-02 via commits cb66c8a, 538718b)	None; direct commits to develop
Fatema Akter Lamia (Developer / reviewer)	Reviewed TC specs & Qodo output	Collaborated on test suite validation & schema setup	Reviewed DEF-01 security audit & DEF-02 backup logic	Co-reviewed defect fixes & verified local execution	None; reviewed direct commits on develop

13. Sign-off and Declaration

We declare that this report describes testing we carried out on the stated version of our software. All results are reproducible from the repository. All tool assistance is disclosed in §10.

	Name	Signature	Date
Prepared by	Faria Khanam Jerin	Faria Khanam Jerin	27/10/2026
Reviewed by	Fatema Akter Lamia	Fatema Akter Lamia	27/10/2026

Appendix A — Evidence Index

#	Evidence	File / URL	Supports section
A1	CI run (tagged commit)	https://github.com/FariaJerin/CityRemedy https://github.com/FariaJerin/CityRemedy/actions	§6.2
A2	Coverage report	docs/testing/evidence/coverage/index.html (HTML export; PHPUnit 13.4.3, php-code-coverage 14.4.1, 26 Oct 2026 20:23 UTC)	§7
A3	Performance test summary	Not produced in v1.0 (deferred)	§8.1
A4	ZAP / Snyk / audit reports	Not produced in v1.0 (deferred)	§8.2
A5	SonarCloud quality gate	Not produced in v1.0 (deferred)	§8.3
A6	Lighthouse / axe reports	Not produced in v1.0 (deferred)	§8.4
A7	Screenshots	docs/testing/evidence/test-output.png; docs/testing/evidence/coverage/COVERAGE-output.png; Figures D1–D4 below	§6, §7
A8	Test source code	tests/Unit (3 files), tests/Integration (5 files); Appendix C	§5
A9	Tool usage evidence	USAGE_LOG.md; branch origin/qodo-test-branch (not merged)	§10
A10	Fix history	Commits cb66c8a (DEF-01), 538718b (DEF-02, parse_feedback_rating)	§9

Appendix B — Pre-Submission Checklist

☒ Commit SHA stated (2ea466a), and all results come from it
☒ Test suite runs from the documented commands (README to be aligned)
☒ CI run is green on the tested commit; link included
☒ Every FR appears in the RTM with ≥1 test case (NFR gaps stated)
☒ Every P1 test case has a detailed specification (§5.2)
☒ Coverage figures match the attached report (81.40% / 84.09%)
☒ No open S1 defects
☒ Every fixed defect has a regression test
☒ All tool-assisted tests are logged in §10
☒ Tool-assisted tests are marked in the code
☒ No real personal data or secrets in test fixtures or screenshots

☒ Template hint text removed; deferred rows justified

☒ Exported as Team10_Exp9.pdf with the Lab Report Cover as page 1

Appendix C — Test-to-Code Traceability

Each test ID maps to one PHPUnit method. Assertion strength is the reviewer's judgement (see §7).

TC ID	File	PHPUnit method	Assertion strength
TC-01	Unit/RoutingTest.php	test_TC01_category_maps_to_correct_department	Strong
TC-02	Integration/FeedbackTest.php	test_feedback_accepts_minimum_and_maximum_ratings	Strong
TC-03	Integration/FeedbackTest.php	test_feedback_rejects_out_of_range_ratings	Strong
TC-04	Integration/FeedbackTest.php	test_feedback_rejects_non_integer_ratings	Strong
TC-05	Integration/ReportTest.php	test_TC05_valid_report_submission_and_lookup	Medium
TC-06	Integration/ReportTest.php	test_TC06_similar_complaints_short_location_returns_empty	Medium
TC-07	Integration/ReportTest.php	test_TC07_public_resolutions_and_timeline	Weak
TC-08	Integration/ReportTest.php	test_TC08_public_stats_aggregation	Weak
TC-09	Integration/ReportTest.php	test_TC09_department_complaint_rows_filtering	Weak
TC-10	Integration/ReportTest.php	test_TC10_admin_metrics_by_role	Weak
TC-11	Integration/FeedbackTest.php	test_feedback_blocked_unresolved_or_wrong_user	Placeholder
TC-12	Integration/AuthTest.php	test_current_user_helper	Strong
TC-13	Integration/AuthTest.php	test_role_check_helpers	Strong
TC-14	Integration/AuthTest.php	test_require_auth_helpers	Medium

TC ID	File	PHPUnit method	Assertion strength
TC-15	Integration/AuthTest.php	test_department_access_permissions	Strong
TC-16	Integration/AuthTest.php	test_login_and_logout_user	Strong
TC-17	Unit/SecurityTest.php	test_csrf_token_generation	Strong
TC-18	Unit/SecurityTest.php	test_csrf_field_generation	Strong
TC-19	Unit/SecurityTest.php	test_csrf_verification_branches	Medium
TC-20	Unit/HelpersTest.php	test_html_escaping	Strong
TC-21	Unit/HelpersTest.php	test_url_helpers	Strong
TC-22	Unit/HelpersTest.php	test_safe_return_to	Strong
TC-23	Unit/HelpersTest.php	test_status_class_mappings	Strong
TC-24	Unit/HelpersTest.php	test_pretty_date_formatting	Strong
TC-25	Unit/HelpersTest.php	test_input_string_helpers	Strong
TC-26	Unit/HelpersTest.php	test_flash_messages_pull	Strong
TC-27	Unit/HelpersTest.php	test_uploaded_file_list_normalization	Strong
TC-28	Unit/HelpersTest.php	test_save_uploaded_file_validation_errors	Strong
TC-29	Unit/HelpersTest.php	test_system_settings_and_logging_helpers	Weak
TC-30	Integration/WorkflowTest.php	test_report_statuses_have_their_expected_display_classes	Redundant
TC-31	Integration/WorkflowTest.php	test_unknown_report_status_uses_neutral_class	Medium
TC-32	Integration/SystemTest.php	test_database_factory_is_loaded	Weak

TC ID	File	PHPUnit method	Assertion strength
TC-33	Integration/SystemTest.php	test_application_base_url_is_configured	Weak
TC-34	Integration/SystemTest.php	test_database_connection_when_mysql_driver_is_available	Medium
TC-35	Integration/SystemTest.php	test_application_name_is_configured	Weak
PT-01	Integration/SystemTest.php	test_PT01_endpoint_response_time	Weak

Appendix D — Evidence Screenshots

Local runs on Windows 11, 27 Oct 2026 (+06), PowerShell in VS Code, branch develop. Coverage was produced with $env:XDEBUG_MODE="coverage".

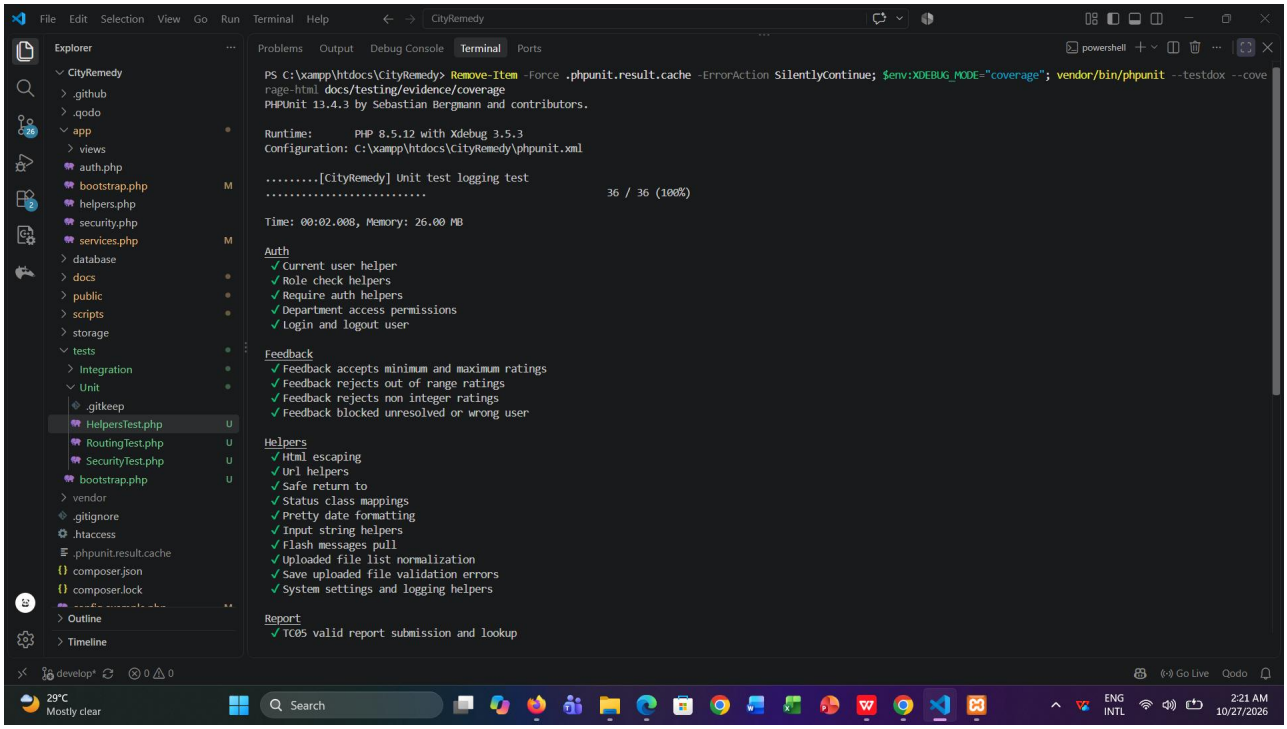


Figure D1 — Test run with --testdox (start): Auth, Feedback, Helpers, Report groups, 02:21

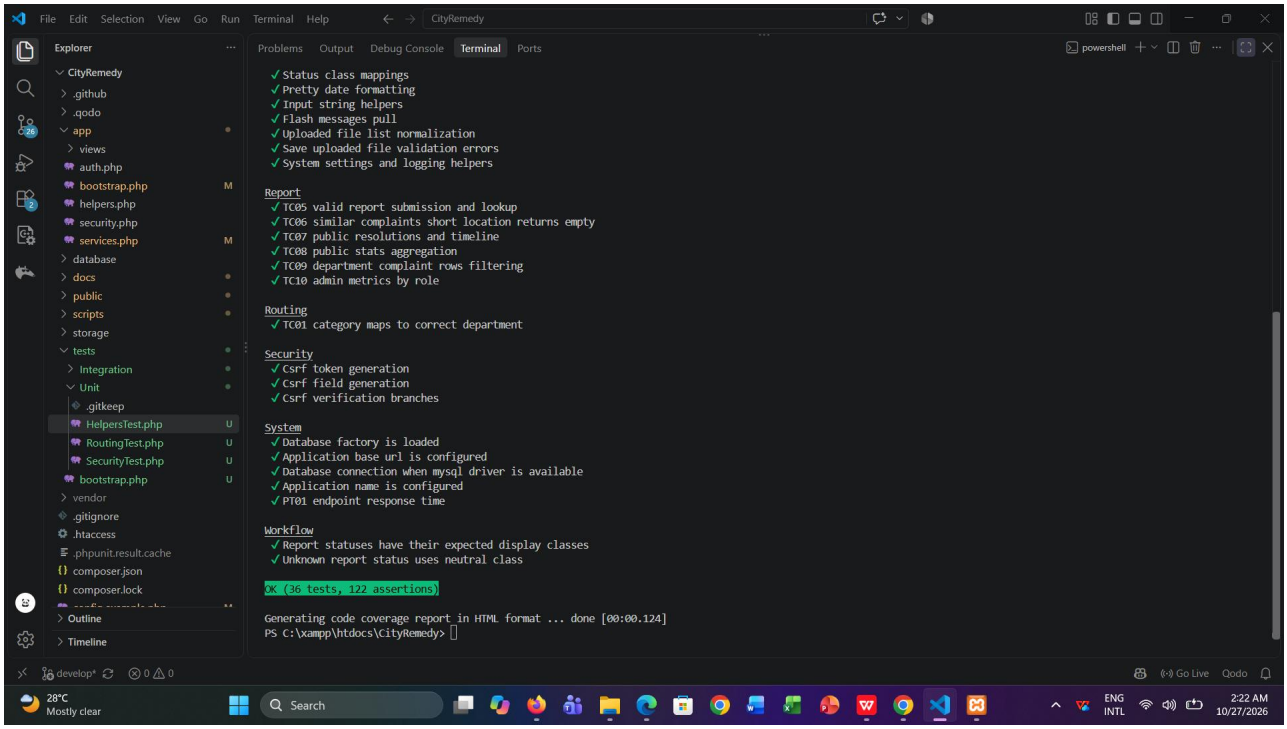


Figure D2 — Test run with --testdox (end): Report, Routing, Security, System, Workflow; OK (36 tests, 122 assertions), 02:22

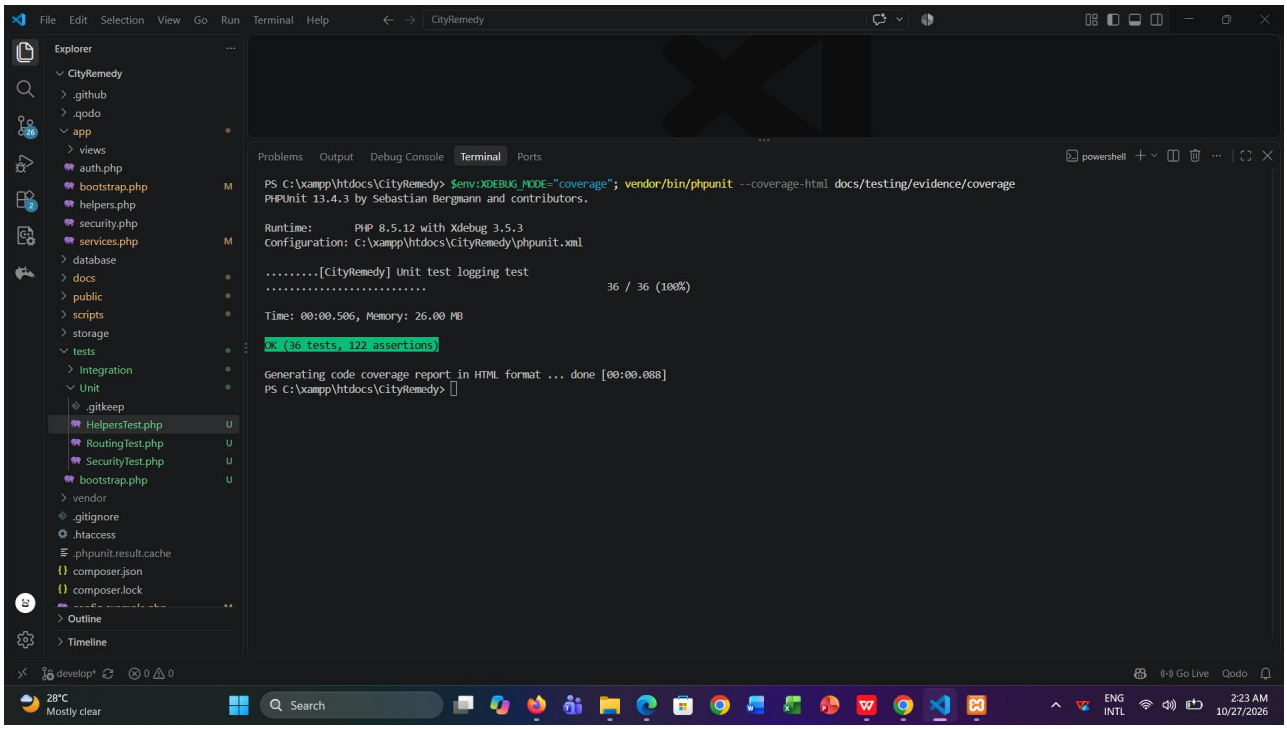


Figure D3 — Second run with coverage generation: 36 / 36 (100%), OK (36 tests, 122 assertions), 02:23

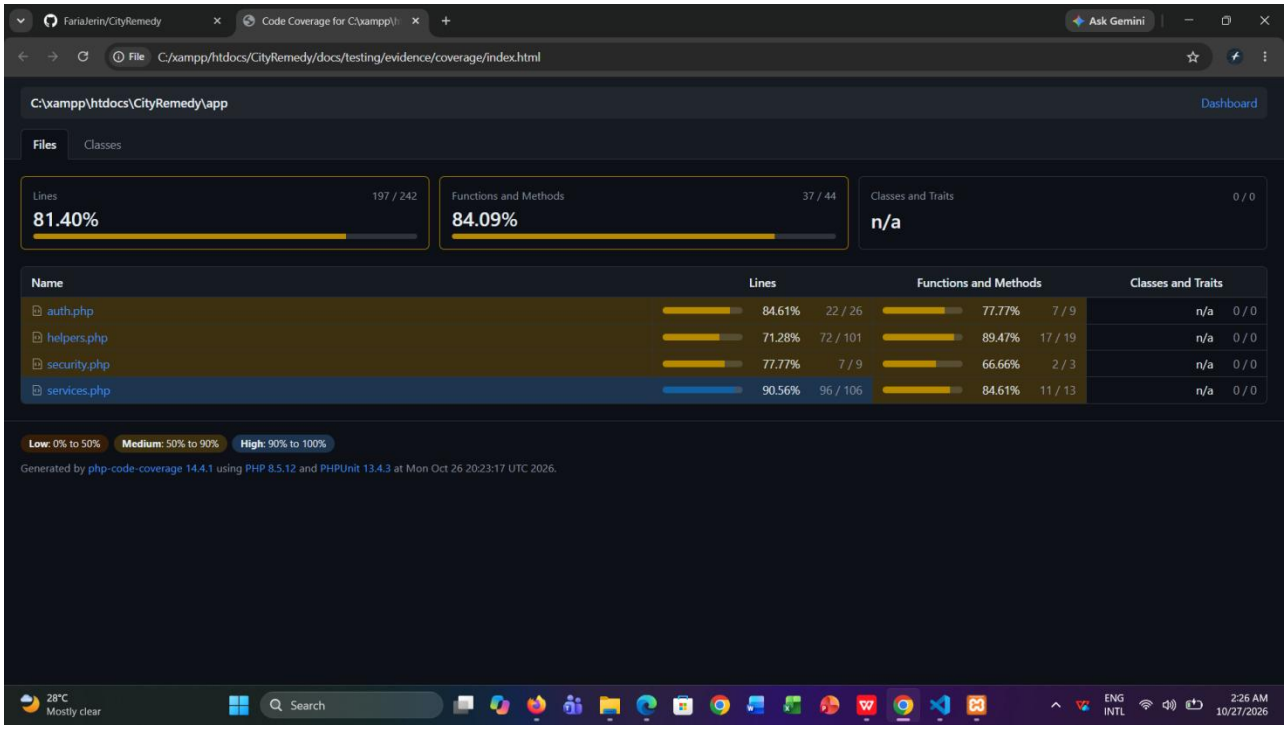


Figure D4 — HTML coverage report: lines 81.40% (197/242), functions 84.09% (37/44), 02:26

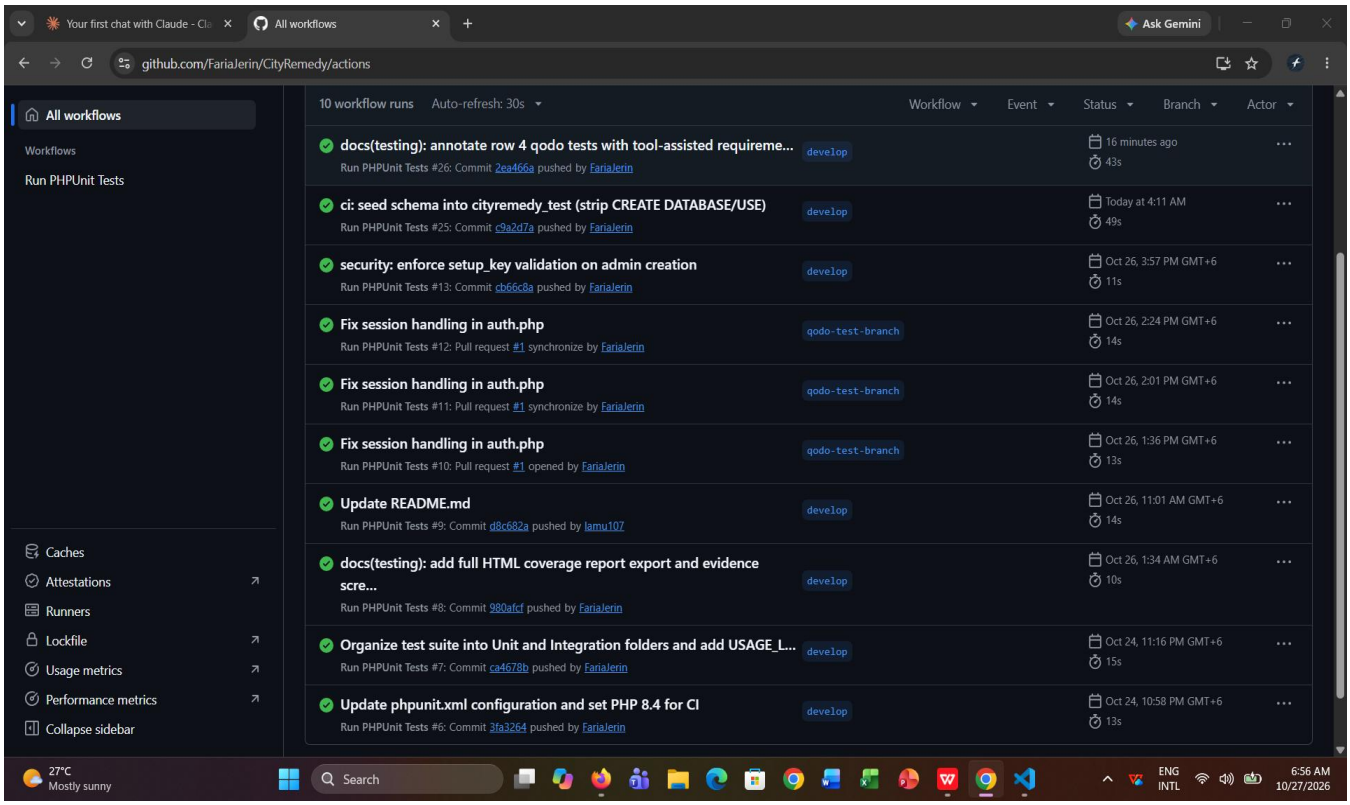


Figure D4 — https://github.com/FariaJerin/CityRemedy/actions